

Molecular mechanisms of curcumin action: gene expression.



Curcumin derived from the tropical plant *Curcuma longa* has a long history of use as a dietary agent, food preservative, and in traditional Asian medicine. It has been used for centuries to treat biliary disorders, anorexia, cough, diabetic wounds, hepatic disorders, rheumatism, and sinusitis. The preventive and therapeutic properties of curcumin are associated with its antioxidant, anti-inflammatory, and anticancer properties. Extensive research over several decades has attempted to identify the molecular mechanisms of curcumin action. Curcumin modulates numerous molecular targets by altering their gene expression, signaling pathways, or through direct interaction. Curcumin regulates the expression of inflammatory cytokines (e.g., TNF, IL-1), growth factors (e.g., VEGF, EGF, FGF), growth factor receptors (e.g., EGFR, HER-2, AR), enzymes (e.g., COX-2, LOX, MMP9, MAPK, mTOR, Akt), adhesion molecules (e.g., ELAM-1, ICAM-1, VCAM-1), apoptosis related proteins (e.g., Bcl-2, caspases, DR, Fas), and cell cycle proteins (e.g., cyclin D1). Curcumin modulates the activity of several transcription factors (e.g., NF- κ B, AP-1, STAT) and their signaling pathways. Based on its ability to affect multiple targets, curcumin has the potential for the prevention and treatment of various diseases including cancers, arthritis, allergies, atherosclerosis, aging, neurodegenerative disease, hepatic disorders, obesity, diabetes, psoriasis, and autoimmune diseases. This review summarizes the molecular mechanisms of modulation of gene expression by curcumin. Copyright © 2012 International Union of Biochemistry and Molecular Biology, Inc.